

An Exact Neumann-Mode Turing Atlas for the Schnakenberg Reaction–Diffusion System

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Abstract

For the Schnakenberg system on a finite interval we derive the unique positive constant equilibrium, its kinetic stability chamber, and the exact two-by-two dispersion matrix for every Neumann mode. Inside the kinetically stable chamber we prove an if-and-only-if continuous wavenumber window, an if-and-only-if finite-domain criterion, an exact strict integer count of unstable modes, and every domain-length entry and exit wall. Compact-resolvent closure and the homogeneous, equal-diffusion, double-wall, endpoint, and excluded zero-diffusion faces complete the linear theorem. Finite exact receipts test the formulas but do not replace their proof. All Route-A gates fail and Route B remains locked.

1 Frozen model and kinetic chamber

Let $a, b, d_u, d_v, L > 0$ and consider, on $0 < x < L$,

$$\begin{aligned} u_t &= d_u u_{xx} + a - u + u^2 v, & u_x &= v_x = 0 \quad \text{at } x = 0, L, \\ v_t &= d_v v_{xx} + b - u^2 v, \end{aligned} \tag{1}$$

Put $s = a + b$. A positive spatially homogeneous equilibrium must satisfy $u^2 v = b$ and $a - u + b = 0$. Thus the unique positive spatially homogeneous equilibrium is

$$(u_*, v_*) = (s, b/s^2). \tag{2}$$

The reaction Jacobian there is

$$J = \begin{pmatrix} (b-a)/s & s^2 \\ -2b/s & -s^2 \end{pmatrix}, \quad \tau := \text{tr } J = \frac{b-a}{s} - s^2, \quad \det J = s^2. \tag{3}$$

Thus the kinetic equilibrium is asymptotically stable exactly when $\tau < 0$. At $\tau = 0$ its eigenvalues are $\pm is$; for $\tau > 0$ it has positive spectral abscissa.

Expand perturbations in the normalized Neumann cosine basis. With

$$\mu_n = (n\pi/L)^2, \quad n = 0, 1, 2, \dots, \tag{4}$$

the n th coefficient evolves under

$$M(\mu) = J - \begin{pmatrix} d_u \mu & 0 \\ 0 & d_v \mu \end{pmatrix}. \tag{5}$$

Direct expansion gives

$$T(\mu) = \text{tr } M(\mu) = \tau - (d_u + d_v)\mu, \tag{6}$$

$$D(\mu) = \det M(\mu) = d_u d_v \mu^2 - B\mu + s^2, \quad B := d_v \frac{b-a}{s} - d_u s^2. \tag{7}$$

Consequently the complete modal pair is

$$\lambda_{n,\pm} = \frac{T(\mu_n) \pm \sqrt{T(\mu_n)^2 - 4D(\mu_n)}}{2}. \tag{8}$$

The phrase *kinetic and modal dispersion closure* marks the baseline manuscript: no continuum-wavenumber test has yet been promoted to a finite interval claim.

2 Continuous window and exact finite-domain selection

Define

$$Q = B^2 - 4d_u d_v s^2. \quad (9)$$

Theorem 1 (Complete linear Turing criterion). *Assume $\tau < 0$. A nonempty open interval of unstable positive squared wavenumbers exists if and only if*

$$B > 0, \quad Q > 0. \quad (10)$$

When these inequalities hold, set

$$\mu_{\pm} = \frac{B \pm \sqrt{Q}}{2d_u d_v}. \quad (11)$$

Then $0 < \mu_- < \mu_+$ and the finite Neumann interval has a genuine linear Turing instability if and only if some $n \geq 1$ satisfies

$$\mu_- < \mu_n < \mu_+. \quad (12)$$

Writing $r_{\pm} = (L/\pi)\sqrt{\mu_{\pm}}$, the exact number of unstable modes is

$$N_{\text{unst}} = \max\{0, \lceil r_+ \rceil - \lfloor r_- \rfloor - 1\}. \quad (13)$$

For fixed $n \geq 1$, its precise domain-length chamber is

$$\frac{n\pi}{\sqrt{\mu_+}} < L < \frac{n\pi}{\sqrt{\mu_-}}. \quad (14)$$

Equality at either wall is neutral rather than unstable.

Proof. Because $\tau < 0$, equation (6) is negative for every $\mu \geq 0$. A real two-by-two matrix with negative trace has positive spectral abscissa exactly when its determinant is negative. Indeed, a negative determinant gives two real eigenvalues of opposite signs, whereas a positive determinant and negative trace put both eigenvalues in the open left half-plane.

The polynomial $D(\mu)$ has positive leading and constant coefficients. It is negative somewhere on the positive axis precisely when its vertex is positive and its minimum is negative, which is exactly (10). The two roots are (11); their sum and product are positive, so both are positive and $D < 0$ precisely between them. This proves the continuous criterion and (12) after restricting to the actual Neumann spectrum.

Taking positive square roots of (12) gives $r_- < n < r_+$. The number of positive integers in this strict interval is exactly (13); the ceiling and floor conventions delete integer endpoints. Solving the same strict inequalities for L yields (14). $\square$

At a window endpoint $D = 0$ and $T < 0$, so the two modal eigenvalues are 0 and T : the zero eigenvalue is simple. The phrase *strict interval mode-count closure* marks the first substantive revision: continuous dispersion, discrete mode selection, multiplicity, and length walls now form one if-and-only-if theorem.

3 Operator closure and all frozen boundaries

Let

$$A = \text{diag}(d_u \partial_{xx}, d_v \partial_{xx}) + J, \quad \mathcal{D}(A) = H_N^2(0, L)^2 \quad (15)$$

on $L^2((0, L); \mathbb{R}^2)$. The diagonal Neumann diffusion operator has compact resolvent; the constant matrix J is bounded. Hence A also has compact resolvent. After complexification, the cosine basis gives an orthogonal direct sum of the blocks $M(\mu_n)$, so (8) exhausts its spectrum with algebraic multiplicity. Since $\mu_n \rightarrow \infty$, no further spectral component is hidden beyond the modal list.

The boundary atlas is exact.

- The homogeneous mode $n = 0$ is precisely J , so a Turing claim always requires kinetic stability and a distinct unstable mode $n \geq 1$.
- If $d_u = d_v = d$, then $B = d\tau < 0$ in the kinetic chamber; equal diffusion cannot create a Turing window.
- If $B > 0$ and $Q = 0$, then D has a double zero as a polynomial in μ . It never becomes negative. A discrete mode exactly at the contact has a simple zero temporal eigenvalue and a negative companion; there is no open unstable window.

- If $Q > 0$, an equality $\mu_n = \mu_-$ or $\mu_n = \mu_+$ is a neutral tie. Strict inequalities are essential in both (13) and (14).
- The faces $d_u = 0$ or $d_v = 0$ are excluded. They are not obtained by dividing (11) by zero, and no degenerate-parabolic extension is claimed.

Exact rational receipts contain nine chambers, 63 modal rows, and 20 length-wall polynomials. Separate modes land exactly on the lower and upper window endpoints, and each open-window chamber records the strict floor/ceiling count. The receipts verify equilibrium, Jacobian, determinant, state classification, both endpoints, and double walls without floating point. They do not prove the continuum theorem; the argument above does.

The named kinetics trace to Schnakenberg’s 1979 paper [1]; the reaction–diffusion instability lineage traces to Turing’s 1952 paper [2]. These citations establish source and terminology boundaries, not priority for the present finite-interval assembly. Nearby owners treat a Brusselator ODE Hopf normal form, scalar Cahn–Hilliard shells, a nonlocal Kuramoto probability PDE, and scalar Fisher–KPP fronts. None owns this unequal-diffusion Schnakenberg Neumann mode atlas.

All Route-A coordinates are FAIL: there is no arithmetic carrier, primitive orbit ledger, target determinant, analytic target bridge, or natural unitary lift. This is a source-local dissipative PDE calculation. No target arithmetic, local data, Euler factors, root number, automorphy, target zero match, Hilbert–Polya operator, or Route-B invocation is claimed. The phrase *operator, boundary, and route closure* marks the second substantive revision.

Remark 2 (Claim boundary). *The result is a complete linear stability classification of the positive homogeneous equilibrium. It is not a theorem on nonlinear patterned branches, nonlinear global well-posedness, attractors, or pattern selection.*

References

- [1] J. Schnakenberg, *Simple chemical reaction systems with limit cycle behaviour*, Journal of Theoretical Biology 81 (1979), 389–400. doi:10.1016/0022-5193(79)90042-0.
- [2] A. M. Turing, *The chemical basis of morphogenesis*, Philosophical Transactions of the Royal Society B 237 (1952), 37–72. doi:10.1098/rstb.1952.0012.